* Linear optimization [Notes on simplex algorithm] (1)

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Note: References taken from "Sundar Vishwanathan", lecture notes, Simplex algorithm here visits extreme points one by one, instead of the LBFS table method.

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1) * Linear Algebra Basics [Required before understanding the actual implementation of simplex algorithm]

⇒ Basis & dimension of a vector space:

A set of vectors is called linearly Independent if no vector in the set can be expressed as a linear combination of the other vectors.

A basis of a vector space V is a set of vectors that satisfy the following two conditions.

i) The vectors are linearly independent.

ii) The vectors span the entire vector space V.

The no of vectors in a basis, is called the dimension of vector space.

* Cardinality of two bases is the same

⇒ Theorem: Any two bases of finite-dimensional vector space have the same number of vectors.

⇒ Proof: let $B = \{v_1, v_2, v_3, \ldots, v_m\}$

& $C = \{w_1, w_2, w_3, \ldots, w_n\}$

be two bases of the same vector space V.

Since B spans V, every vector in C can be expressed as a linear combination of vectors in B.

Since, if a vector space is spanned by m vectors, then any linearly independent set in that space contains m vectors.

Since C is linearly independent, we get

$$n \leq m$$

Similarly C spans V, & B is LID,

$$\therefore m \leq n$$

Since, $n \leq m$ and $m \leq n$

we have $m = n$

Hence, any two bases of the same vector space have the same cardinality

This ensures that the dimension of a vector space is well defined.

* Row rank equals column Rank!

For a matrix A: ↳ The row rank is the maximum number of linearly independent row.

↳ Similarly, the column rank is no. of LID columns.

* Theorem: rank(A) = row rank(A) = column rank(A)

* Proof: (Using Gaussian Elimination)

Suppose A is a mxn matrix. By applying elementary row operations we can transform A into row echelon form R.

Elementary row operations are invertible, so they preserve the linear dependence relation among rows, consequently, the number of LID rows is the no. of no-zero rows in R.

Let r be the no. of pivots in echelon form.

The pivot ~~cols~~ columns of R are LID, & the non-pivot columns can be expressed in terms of the pivot columns. Thus, the column rank is also r.

$$\therefore \text{rank}(A) = r = \text{Row rank} = \text{Column rank}.$$

In Simplex algorithm, rank tells us how many independent constraints or directions of constraints we have

* Rank-Nullity theorem: let A be a $m \times n$ matrix. The null space of A is:

$$\text{Null}(A) = \{x \in \mathbb{R}^n : Ax = 0\}$$

It's dimension is the nullity of A.
The rank-nullity theorem states that:

$$\text{rank}(A) + \text{nullity}(A) = n.$$

The null space of the tight constraints matrix tell us which direction preserves the currently tight constraints. This is useful when determining whether a point is a vertex & we can move along the edge.

2) Convex Sets & the feasible region.

* Convex sets: A set $S \subseteq \mathbb{R}^n$ is called a convex set, if for any two points $x, y \in S$, the entire line segment joining them also lies in S.

mathematically, for every $\lambda \in [0,1]$

$$\lambda x + (1-\lambda) y \in S$$

* Feasible region in linear programming.

Consider a LPP:

$$\max \; c^T x$$
$$\text{s.t.} \; Ax \leq b$$

Here, $x \in \mathbb{R}^n$ is the vector of variables & each row of $Ax \leq b$ represents a linear equality.
The feasible region is the set of all points satisfying every constraints:

$$P = \{x \in \mathbb{R}^n : Ax \leq b\}$$

Each linear inequality defines a half-space, which is convex. The intersection of convex sets is also convex. Therefore a feasible region of linear program is convex.

* Vertices & tight constraints.

A point x^* is a vertex of a convex set P, if it cannot be written as a non trivial combination of two distincts points of P.

i.e. if, $x^* = \lambda y + (1-\lambda) z$

where, $y, z \in P$ & $0 < \lambda < 1$, then

$$y = z = x^*$$

A constraint $a_i^T x \leq b_i$ is called tight at x^* if

$$a_i^T x^* = b_i$$

Otherwise it is called a slack constraint.

For eg. consider $x+y \leq 4$, $x \leq 2$, $y \leq 3$

At $(2,2)$, the first two constraints are tight

i.e. $2+2 = 4$, $2 = 2$

$y \leq 3$ is not tight $\because 2 < 3$

3) Rank of the tight constraints at a vertex.

Consider the feasible region:

$$P = \{x \in \mathbb{R}^n : Ax \leq b\}$$

let x^* be a feasible point, & let $I(x^*)$ denote the set of constraints that are tight at x^*.

Construct the matrix A_I using the rows of A corresponding to these tight constraints.

Theorem: A feasible point x^* is a vertex iff

$$\text{rank}(A_I) = n$$

This assumes the variables are in $\mathbb{R}^n$ & all relavant constraints are included.

* Proof : If the rank is n, the point is a vertex.

Now we suppose, rank $(A_I) = n$

Assume that x^* is not a vertex. Then there exist distinct feasible point $y, z \in P$ & $0 < \lambda < 1$ such that:

$$x^* = \lambda y + (1-\lambda)x$$

Consider any constraint that is tight at x^*

$$a_i^T x^* = b_i$$

Substituting the convex combination

$$\lambda a_i^T y + (1-\lambda)\, a_i^T z = b_i$$

Such y, z are feasible

$$a_i^T y \leq b_i \quad \& \quad a_i^T z \leq b$$

A weighted average of these two quantities can equal b_i, with both weights strictly positively, iff both quantities equal b_i. Thus,

$$a_i^T y = a_i^T z = b_i$$

This holds for every tight constraint. Consequently

$$A_I (y - z) = 0$$

But A_I has rank n, so its null space contains only null vector

$$\therefore y - z = 0 \Rightarrow y = z$$

This contradicts that y & z are distinct

$$\therefore x^* \text{ is a vertex} \iff \text{rank}(A_I) = n$$

i.e. At a vertex in n-dimensional space, the tight constraints normal span all of $\mathbb{R}^n$. This means there is no nonzero direction that preserves every tight constraint while remaining on both sides of the point.

∴ At vertex movement is restricted in every direction.

The simplex algorithm uses the aforementioned property to move from one vertex to another. To leave a vertex along an edge, we allow one of the currently tight constraints to become slack while keeping enough other independent constraints tight to define the edge.

4) Why optimum occurs at the vertex?

soln:

Consider a LPP:

$$\max_{x \in P} c^T x$$

where P is the feasible region

The objective function is linear. Unlike a quadratic function, it's value changes at a constant rate along any straight line.

* Objective value along a line segment.

let $x, y \in P$ & consider a point on the line segment joining them.

$$z = \lambda x + (1-\lambda) y \quad , \quad 0 \leq \lambda \leq 1$$

The objective value at z is

$$c^T z = c^T (\lambda x + (1-\lambda) y)$$

$$c^T z = \lambda c^T x + (1-\lambda) c^T y$$

The objective value at an intermediate point is simply a weighted average of it's values at the end points.

Therefore, if one endpoint has a higher objective value, the interior of the segment cannot have a value higher than that endpoint.

If both endpoints have the same objective value, every point on the segment has that same value.

* Searching for the optimum among the vertices

Since our feasible region is a polygonal area with a finite number of vertices when it has finitely many defining inequality & it's pointed. A standard LPP feasible region with $x \geq 0$ is pointed.

For a non-empty, pointed polygonal feasible region, if a linear objective has a finite attained optimum, at least one optimal vertex exists.

The intitution is that a non-vertex point lies in a line segment within the feasible region. By moving along that segment in a direction that doesn't decrease the objective, we can reach an endpoint without worsening the objective. Repeating this process leads to a vertex with an equally good or a better objective value.

Thus, we do not need to search every point of the feasible region & we can just look at vertices instead.

5) Moving from one vertex to another with an increase in the objective. (central idea of the simplex algorithm).

⇒ Suppose we are at a feasible vertex x.
let the objective function be:

$$f(x) = c^T x.$$

we want to find a direction d such that moving from x in the direction d.

⇒ is inside the feasible region.
⇒ increases the objective function.
⇒ takes us to another vertex.

* Choosing the feasible direction

Consider the constraint:

$$a_i^T x \leq b_i$$

if constraint i is tight at x, then

$$a_i^T x = b$$

To move along a direction d, while keeping this constraint tight, we require:

$$a_i^T d = 0$$

∴ if we want to move along an edge, we choose a direction that preserves the constraint defining that edge.

In an n-dimensional, non degenerate situation, an edge is typically determined by $n-1$ independent tight constraints. A direction along that edge satisfies

$$A_j d = 0$$

where A_j contains the $n-1$ independent rows defining the edge.

By the rank-nullity theorem, if A_j has rank $n-1$ it's null space has dimension

$$n-(n-1) = 1$$

Hence, the direction along the edge is determined upto a scalar multiple & a choice of sign.

For a degenerate vertex, more than n constraints may be tight, so we choose an appropriate, independent subset to describe an candidate edge.

* Checking if the objective increases.

The new objective function at new point $x + td$ is:

$$f(x+td) = c^T(x+cd)$$
$$= c^T x + tc^T d$$

$$\therefore \quad f(x+td) - f(x) = tc^T d$$

For $t > 0$, the objective fn increases if

$$c^T d > 0$$

i.e.

if $c^T d > 0$, moving along d increases the obj

if $c^T d = 0$, objective remains same

if $c^T d < 0$, objective decreases, so not useful.

* moving along d to reach optimal.

Even if a direction improves the objective we cannot move indefinitely because we must remain feasible.

Suppose current x, satisfies : $Ax \leq b$

moving to $x + td$: $A(x + td) \leq b$

i.e. $Ax + Atd \leq b$

For each constraint i,

$$a_i^T x + t a_i^T d \leq b_i$$

if $a_i^T d > 0$, the constraint places an upper bound on t:

$$t \leq \min_{i : a_i^T d > 0} \frac{b_i - a_i^T x}{a_i^T d}$$

At the limiting step, at least one previously inactive constraint becomes tight. The current vertex reaches the boundary of its feasible movement & the endpoint is typically an adjacent vertex.

If no constraint imposes an upper bound on t, the ray remains feasible indefinitely. If $c^T d > 0$ the objective is unbounded.

Thus, in conclusion: to move along an edge

1. One of the current tight constraints is released
2. A direction is chosen that preserves the remaining independent tight constraints.
3. The movement continues until the another constraint becomes tight.
4. The new tight constraint replaces the released constraint in the description of the vertex.

* The simplex algorithm: This using the ideas of convexity, rank, tight constraints & moving along the edges.

Consider the LPP:

$$\max \quad z = c^T x$$
$$\text{s.t.} \quad Ax \leq b$$

Assume that given a feasible starting vertex $x(0)$, our aim is to move from one vertex to another, while increasing the objective value until we reach an optimal vertex.

Algorithm:

Step 1: Start at a vertex

let the current vertex be x. Calculate the objective value.

$$z = c^T x$$

Step 2: Identify the tight constraints

Find all constraints that satisfy

$$a_i^T x = b_i$$

Construct the matrix A_I using the correspondiy rows of A. At a vertex in n-dimensional space, these rows must have rank n!

$$\text{rank}(A_I) = n.$$

Step 3: Find possible edge directions:

Choose a suitable set of independent tight constraints and find a direction d that preserves the constraints defining an edge.

For a non degenerate vertex in n-dimensional space, an edge is typically defined by n-1 independent tight constraints, therefore, the direction satisfy

$$A_j d = 0$$

where A_j contains the n-1 rows correspond to these constraints.

Step 4: Check whether the objective increases.

Moving from x in direction d gives the new point

$$x_{new} = x + td$$

where $t \geq 0$, is the step length.

The change in objective value is

$$c^T(x+td) - c^Tx = tc^Td$$

Hence for $t > 0$, the obj increases if

$$c^Td > 0$$

Consider only feasible directions satisfying this condition.

Step 5: Find the max feasible step length

we must ensure that new point satisfies all the constraint.

$$A(x+td) \leq b$$

For each constraint $a_i^Td > 0$, we obtain

$$t \leq \frac{b_i - a_i^Tx}{a_i^Td}$$

∴ The max feasible step length is

$$t_{max} = \min_{i:\, a_i^Td > 0} \frac{b_i - a_i^Tx}{a_i^Td}$$

Step 6: Move onto the next vertex

Update the current point:

$$x_{new} = x + t_{max}d$$

At the limiting step, at least one additional constraint becomes tight. Update the set of tight constraints & verify that their normal vectors have rank n.

Step 7: Check for optimality & repeat

If no feasible adjacent edge provides a direction that increases the objective, the current vertex is optimal.

otherwise, set $x = x_{new}$ & repeat from step 2.

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* E.g. Consider the problem:

Max: $z = 3x + 2y$

s.t. $x + y \leq 4$

$x \leq 2$

$y \leq 3$

$x, y \geq 0$

Assume an initial vertex $(0,0)$.

Step 1: move from $(0,0)$ to $(2,0)$

Choose the direction: $d = (1,0)$

Then, $c^T d = (3\ 2)\begin{pmatrix}1\\0\end{pmatrix} = 3 > 0$

The constraint $x \leq 2$ limits the step length to $t = 2$.

The constraint $x + y \leq 4$, would allow a step length of 4.

$\therefore$ we move to $(x,y) = (0,0) + 2(1,0) = (2,0)$

Then, $z = 3(2) + 2(0) = 6$

Step 2: move from $(2,0)$ to $(2,2)$

At $(2,0)$ the constraints are $x = 2$ & $y = 0$ are tight

Choose $d = (0,1)$

Then, $c^T d = (3\ 2)\begin{pmatrix}0\\1\end{pmatrix} = 2 > 0$,

Again, $(x,y) = (2,0) + 2(0,1) = (2,2)$

The obj. value becomes: $z = 3(2) + 2(2) = 10$

Step 3: check whether $(2,2)$ is optimal

$x = 2$, $x + y = 4$ are tight

The normal vectors are $(1,0)$ & $(1,1)$ which are LID

$\therefore$ Their matrix has rank $= 2$, $\therefore$ $(2,2)$ is a vertex.

$\therefore$ max $z = 10$ at $(x,y) = (2,2)$